

Summary: Post-Market Monitoring for AI Systems

1. Introduction & The Paradigm Shift

Traditional product safety frameworks assume that a product evaluated as compliant at market entry stays compliant throughout its lifecycle (e.g., stable chemical formulations in pharma, or predictable degradation in hardware). Artificial Intelligence systems completely break this assumption.

Due to continuous fine-tuning, online learning, or changes in real-world data distributions, an AI system's performance profile naturally shifts over time. This necessitates a dynamic regulatory response inspired by, yet more complex than, pharmaceutical **pharmacovigilance** and medical device surveillance.

2. Core Legal Obligations Under the EU AI Act

The paper outlines a heavily prescriptive framework taking effect on **2 August 2026**:

- **Article 72 (The Primary Provider Obligation):** Providers of high-risk AI systems must establish a proactive, systematic post-market monitoring system *before* the system hits the market. It cannot be retrofitted post-deployment.
- **Annex IV (Technical Documentation):** Requires a living **Post-Market Monitoring Plan** that strictly defines performance metrics (accuracy, robustness, cybersecurity), data collection frequencies, and specific alert thresholds.
- **Article 26(5) (Deployer Obligations):** Deployers are legally bound to monitor systems based on the provider's instructions, maintain automated logs, and immediately flag risks or non-conformity to the provider.
- **Article 55 (GPAI Systemic Risk Monitoring):** Providers of General-Purpose AI models with systemic risk ($>10^{25}$ FLOPs) face unique, model-level obligations, including mandatory annual adversarial testing and tracking downstream incident signals across highly varied applications.

3. Technical Bottlenecks in AI Monitoring

Clark identifies four major technical hurdles that separate AI monitoring from standard IT infrastructure surveillance:

A. Model Drift Taxonomy & Detection

- **Covariate Shift:** Input data distribution changes, but the input-to-output relationship remains stable.
- **Label Shift:** The distribution of the model's predicted classes changes over time.
- **Concept Drift:** The fundamental statistical relationship between inputs and outputs changes.
- *Detection Tools:* Providers must document technically sound choices using tools like Population Stability Index (PSI), Kolmogorov-Smirnov tests, or sequential alerting frameworks like CUSUM and ADWIN.

B. Latent Ground Truth

Unlike clinical events in pharma that happen concurrently with drug usage, an AI system's correctness often has vast latency. For example, a credit-scoring model's accuracy can only be fully verified years later when loans mature. Monitoring systems must rely on indirect proxies rather than direct accuracy measurements.

C. Fairness Metric Instability

AI systems that are fair at launch can develop demographic disparities as populations shift. Providers must monitor disaggregated data across protected groups. However, mathematically, it is impossible to satisfy all fairness definitions (e.g., demographic parity vs. calibration) simultaneously, creating a strict normative and legal trade-off.

D. Adversarial Robustness & Explainability

Systems face active strategic manipulation (e.g., users gaming a credit model or data poisoning sensor feeds). Monitoring requires continuous anomaly detection and periodic red-teaming. Furthermore, shifts or decay in explanation models (like SHAP or LIME) can invalidate human oversight mechanisms under Article 14.

4. The Grand Legal Conflicts (The "Regulatory Intersection")

The paper's core legal contribution highlights critical design conflicts where compliance with one regulation risks violating another.

Conflict 1: Article 72 AI Act vs. Article 5(1)(e) GDPR (Storage Limitation)

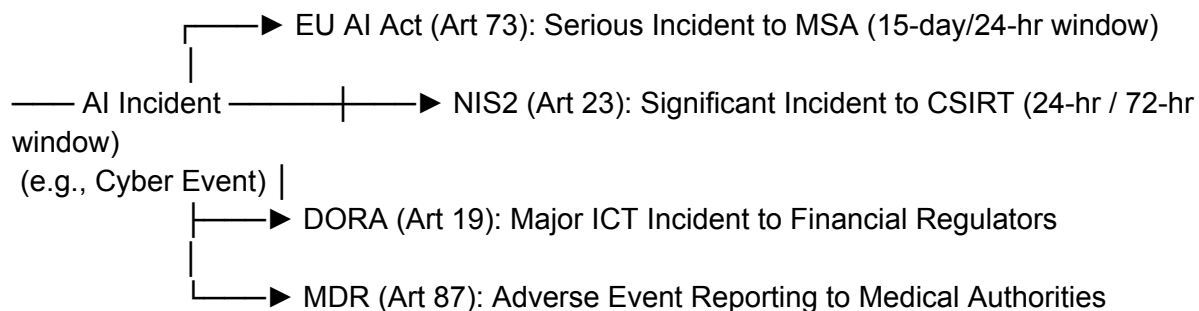
EU AI Act (Article 72)	GDPR (Article 5(1)(e))

Mandates <i>active and systematic collection and retention</i> of system performance data throughout its operational lifetime.	Requires that personal data be kept <i>no longer than necessary</i> for the original purpose.
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The Issue: If monitoring logs contain patient data or financial histories, keeping them for years to monitor statistical drift directly clashes with data protection rules. Clark notes that as of April 2026, neither the EDPB nor the AI Office has resolved this dispute.

Conflict 2: Overlapping Incident Reporting Regimes

A single AI operational failure can simultaneously trigger multiple independent reporting channels with completely different timelines and taxonomies:



5. Proposed Unified Implementation Architecture

To bridge the gap between abstract compliance plans and engineering realities, Clark argues that the Post-Market Monitoring Plan must be treated as a **core design artifact** rather than a retrospective paperwork exercise:

1. **Privacy-by-Design Data Pipelines:** Resolve the GDPR tension at the collection point (ingesting aggregate statistical summaries, using sampling strategies, or inserting differential privacy noise *before* storage) rather than relying on a retention policy.
2. **Three-Tier Threshold Alerting:** Calibrate Statistical Process Control charts (like CUSUM) to separate internal review anomalies from systemic degradation alerts, and isolate true regulatory "serious incidents" to avoid toxic over- or under-reporting.
3. **Unified Governance Architecture:** Implement a single organizational taxonomy so a single root incident record can automatically auto-populate customized evidence packages required for the AI Office, CSIRTs, and financial/medical auditors simultaneously.